

semantic Interpretation and similarity

using TF-IDF and ESA

Step-by-Step process	Page
1. Problem Document Collection	01
2. Vocabulary Construction	01
3. Term Frequency (TF) Matrix	01
4. Document Frequency (DF) and (IDF) calculation	02
5. Calculate Inverse Document Frequency (IDF)	02
(i) Document frequency (df) (ii) IDF formula (iii) IDF values	
6. TF-IDF Matrix	03
7. ESA vector representation	04
8. ESA vector for "green" crocodile	05
9. ESA vector for "big" crocodile	06
10. ESA vector for "scary" crocodile	06
11. cosine similarity computation	07-08
12. Final Answer	09

Document Collection:

d_1 : "scary green crocodile"

d_2 : "scary green big"

d_3 : "small crocodile"

① Find Unique Words (Vocabulary)

Arrange Words alphabetically

vocabulary = (big, crocodile, green, scary, small)

② Calculate Term Frequency (TF) matrix

Find how many times each unique word occurs in a document

Term \ Doc ↓	big	crocodile	green	scary	small
d_1	0	1	1	1	0
d_2	1	0	1	1	0
d_3	0	1	0	0	1

③ Calculate document frequency (df) and IDF :

df = Find how many times each unique word occurs in full document collection.

big = 1, crocodile = 2, green = 2

scary = 2 and small = 1

Calculate Inverse Document Frequency (IDF)

Calculate inverse document frequency (IDF), which measures how important a word is using the formula :-

$$\text{idf}(\text{term}) = \log_2 \frac{N}{\text{df}(\text{term})}$$

Where, $N=3$, number of documents.

For "big" ∴ $\text{df}["\text{big}"] = 1$

$$\text{idf}["\text{big}"] = \log_2 \left(\frac{3}{1} \right) = 1.585$$

For "crocodile" ∴ $\text{df}["\text{crocodile}"] = 2$

$$\text{idf}["\text{crocodile}"] = \log_2 \left(\frac{3}{2} \right) = 0.585$$

For "green" ∴ $\text{df}["\text{green}"] = 2$

$$\text{idf}["\text{green}"] = \log_2 \left(\frac{3}{2} \right) = 0.585$$

For "scary" ∴ $\text{df}["\text{scary}"] = 2$

$$\text{idf}["\text{scary}"] = \log_2 \left(\frac{3}{2} \right) = 0.585$$

For "small" ∴ $\text{df}["\text{small}"] = 1$

$$\text{idf}["\text{small}"] = \log_2 \left(\frac{3}{1} \right) = 1.585$$

(3)

Derived df and idf matrix

Term	df	idf
big	1	1.585
crocodile	2	0.585
green	2	0.585
seamy	2	0.585
small	1	1.585

④ Calculate TF-IDF (Term-concept matrix) :-

$$tfidf(i,j) = tf(i,j) \times idf(i)$$

where, $tf(i,j)$ = frequency of term t_i in document d_j .

$idf(i)$ = inverted document frequency of term i .

$$tfidf = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1.585 \\ 0.585 \\ 0.585 \\ 0.585 \\ 1.585 \end{bmatrix}$$

$$= \begin{matrix} 0 \times 1.585 & 1 \times 0.585 & 1 \times 0.585 & 1 \times 0.585 & 0 \times 1.585 \\ 1 \times 1.585 & 0 \times 0.585 & 1 \times 0.585 & 1 \times 0.585 & 0 \times 1.585 \\ 0 \times 1.585 & 1 \times 0.585 & 0 \times 0.585 & 0 \times 0.585 & 1 \times 1.585 \end{matrix}$$

$$= \begin{bmatrix} 0 & .585 & .585 & .585 & 0 \\ 1.585 & 0 & .585 & .585 & 0 \\ 0 & .585 & 0 & 0 & 1.585 \end{bmatrix}$$

(4)

Question-1

Calculate semantic interpretation for text
"green crocodile"

① Calculate term-frequency (tf) for "green crocodile"

Term	TF	TF-IDFQ
big	0	0
crocodile	1	.585
green	1	.585
scary	0	0
small	0	0

② Now, calculate term-concept vector using

$$V_Q = \text{tfidf} \times \text{TFIDFQ}$$

$$= \begin{bmatrix} 0 & .585 & .585 & .585 & 0 \\ 1 & .585 & 0 & .585 & 0 \\ 0 & .585 & .585 & 0 & 1.585 \end{bmatrix} \times \begin{bmatrix} 0. \\ .585 \\ .585 \\ 0 \\ 0 \end{bmatrix}$$

For document d_1 :-

$$0 \times 0 + .585 \times .585 + .585 \times .585 + .585 \times 0 + 0 \times 0$$

$$= 0.342 + 0.342$$

$$= 0.684$$

⑤

For document d_2 :-

$$1.585 \times 0 + 0 \times .585 + .585 \times .585 + .585 \times 0 + 0 \times 0 \\ = 0.342$$

For document d_3 :-

$$0 \times 0 + .176 \times .176 + 0 \times .176 + 0 \times 0 + 1.585 \times 0 \\ = 0.342$$

$$\therefore V_Q = \begin{bmatrix} 0.684 \\ 0.342 \\ 0.342 \end{bmatrix}$$

Final Interpretation :-

document	score	meaning
d_1	.684	most related
d_2	.342	less related
d_3	.342	less related

(6)

Question - 02

Calculate similarity between "big crocodile" and "scary crocodile" based on ESA score.

Let $A = \text{"big crocodile"}$

$B = \text{"scary crocodile"}$

For A :-

Term	TF	TF IDF - A
big	1	1.585
crocodile	1	.585
green	0	0
scary	0	0
small	0	0

Now, calculate term - concept vector $V_A = \text{tfidf} \times \text{TFIDFA}$

$$\begin{bmatrix} 0 & .585 & .585 & .585 & 0 \\ 1.585 & 0 & .585 & .585 & 0 \\ 0 & .585 & 0 & 0 & 1.585 \end{bmatrix} \times \begin{bmatrix} 1.585 \\ .585 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore V_A = \begin{bmatrix} 0.342 \\ 2.512 \\ 0.342 \end{bmatrix}$$

For - B

Term	TF	TF IDF - B
big	0	0
crocodile	1	.585
green	0	0
scary	1	.585
small	0	0

Calculate :-

$$V_A = \text{tfidf} \times \text{TF IDF-B}$$

$$V_B = \begin{bmatrix} 0 & .585 & .585 & .585 & 0 \\ 1.585 & 0 & .585 & .585 & 0 \\ 0 & .585 & 0 & 0 & 1.585 \end{bmatrix} \times \begin{bmatrix} .0 \\ .585 \\ 0 \\ .585 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} .684 \\ .342 \\ .342 \end{bmatrix}$$

Next,

calculate the cosine similarity :-

$$\text{cosine similarity (A, B)} = \frac{V_A \cdot V_B}{\|V_A\| \|V_B\|}$$

$$\begin{aligned}
 V_A \cdot V_B &= \begin{bmatrix} 0.342 \\ 2.512 \\ 0.342 \end{bmatrix} \times \begin{bmatrix} 0.684 \\ 0.342 \\ 0.342 \end{bmatrix} \\
 &= 0.342 \times 0.684 + 2.512 \times 0.342 + 0.342 \times 0.342 \\
 &= 0.23392 + 0.8591 + 0.11694 \\
 &= 1.20996
 \end{aligned}$$

Now,

$$\begin{aligned}
 \|V_A\| &= \sqrt{.342^2 + 2.512^2 + .342^2} = \sqrt{.11696 + 6.310 + .11696} \\
 &= \sqrt{6.543928} \\
 &= 2.55811
 \end{aligned}$$

$$\begin{aligned}
 \|V_B\| &= \sqrt{.684^2 + .342^2 + .342^2} = \sqrt{.4678 + .11696 + .11696} \\
 &= \sqrt{0.701728} \\
 &= 0.837692
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Cosine-similarity } (V_A, V_B) &= \frac{V_A \cdot V_B}{\|V_A\| \|V_B\|} \\
 &= \frac{1.20996}{2.55811 \times 0.837692} \\
 &= \frac{1.20996}{2.1429} \\
 &= 0.5646
 \end{aligned}$$

Final Answer:

1. ESA Vector for "green crocodile" = $\begin{bmatrix} .684 \\ .342 \\ .342 \end{bmatrix}$

document	score	meaning
d ₁	.684	most related
d ₂	.342	less related
d ₃	.342	less related

2. Similarity ("big crocodile" "scary crocodile")
= 0.5646

□ A cosine similarity of 0.5646 in ESA indicates a moderate semantic relatedness between the two vectors V_A and V_B meaning the corresponding texts share several underlying knowledge concepts but are not semantically identical.